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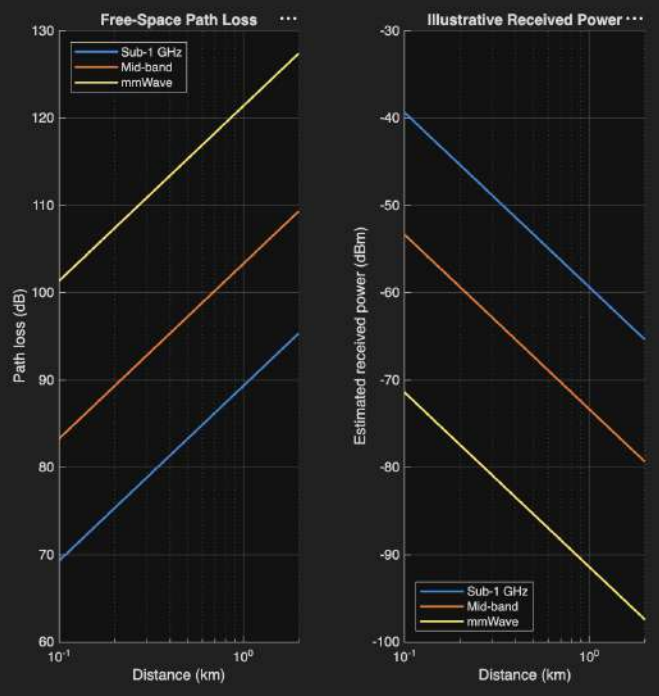
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```
clear; clc; close all;
freqGHz = [0.7 3.5 28];
freqMHz = freqGHz * 1000;
bandName = ["Sub-1 GHz"; "Mid-band"; "mmWave"];
txPowerdBm = 30;
gtDbi = 0; grDbi = 0;
distanceKm = logspace(log10(0.1),log10(2),200);
distanceAtPointKm = 0.5;
fspl = @(fMHz,dKm) 32.44 + 20*log10(fMHz) + 20*log10(dKm);
lossAtPoint = fspl(freqMHz,distanceAtPointKm);
rxAtPoint = txPowerdBm + gtDbi + grDbi - lossAtPoint;
linkTable = table(bandName,freqGHz(:),lossAtPoint(:),rxAtPoint(:), ...
    'VariableNames',{'Band','Frequency_GHz','FSPL_dB','EstimatedRx_dBm'});
disp(linkTable);
figure; tiledlayout(1,2);
nexttile; hold on; grid on;
for k=1:numel(freqMHz)
    plot(distanceKm,fspl(freqMHz(k),distanceKm),'LineWidth',1.5);
end
set(gca,'XScale','log'); xlabel('Distance (km)'); ylabel('Path loss (dB)');
legend(bandName,'Location','northwest'); title('Free-Space Path Loss');
nexttile; hold on; grid on;
for k=1:numel(freqMHz)
    rx = txPowerdBm + gtDbi + grDbi - fspl(freqMHz(k),distanceKm);
    plot(distanceKm,rx,'LineWidth',1.5);
end
set(gca,'XScale','log'); xlabel('Distance (km)'); ylabel('Estimated received power (dBm)');

legend(bandName,'Location','southwest'); title('Illustrative Received Power');
cellRadiusKm = 0.5;
reuseN = [1 3 7 12];
pathLossExponent = 3.5;
reuseDistance = sqrt(3*reuseN) * cellRadiusKm;
sirLinear = (reuseDistance/cellRadiusKm).^pathLossExponent / 6;
sirDb = 10*log10(sirLinear);
reuseTable = table(reuseN(:),reuseDistance(:),sirDb(:), ...
    'VariableNames',{'ReuseClusterN','ApproxInterfererDistance_km','ApproxSIR_dB'});
disp(reuseTable);
figure; plot(reuseN,sirDb,'o-','LineWidth',1.5); grid on;
xlabel('Reuse cluster size N'); ylabel('Approximate SIR (dB)');
title('Simplified Co-Channel SIR versus Reuse Cluster Size');
fprintf('At %.2f km, estimated received powers are:\n',distanceAtPointKm);
disp(table(bandName,rxAtPoint(:),'VariableNames',{'Band','Rx_dBm'}));
```

Figures

Figure 1 | Figure 2



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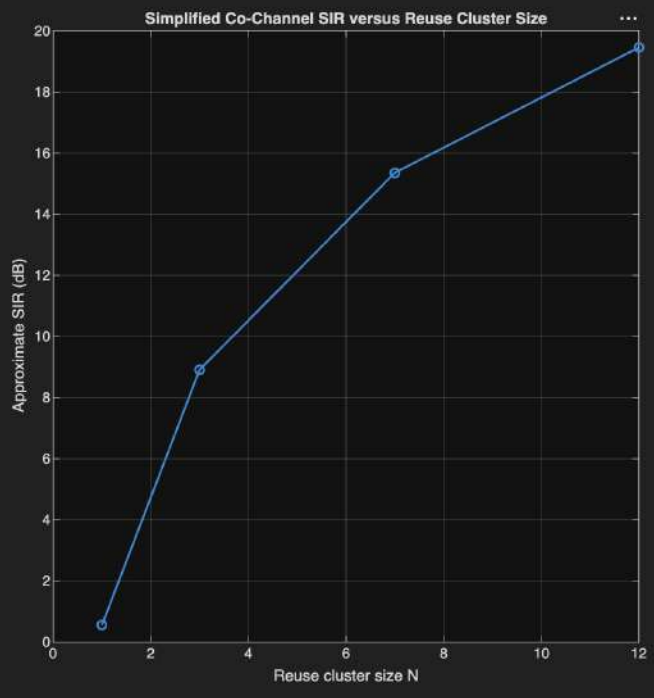
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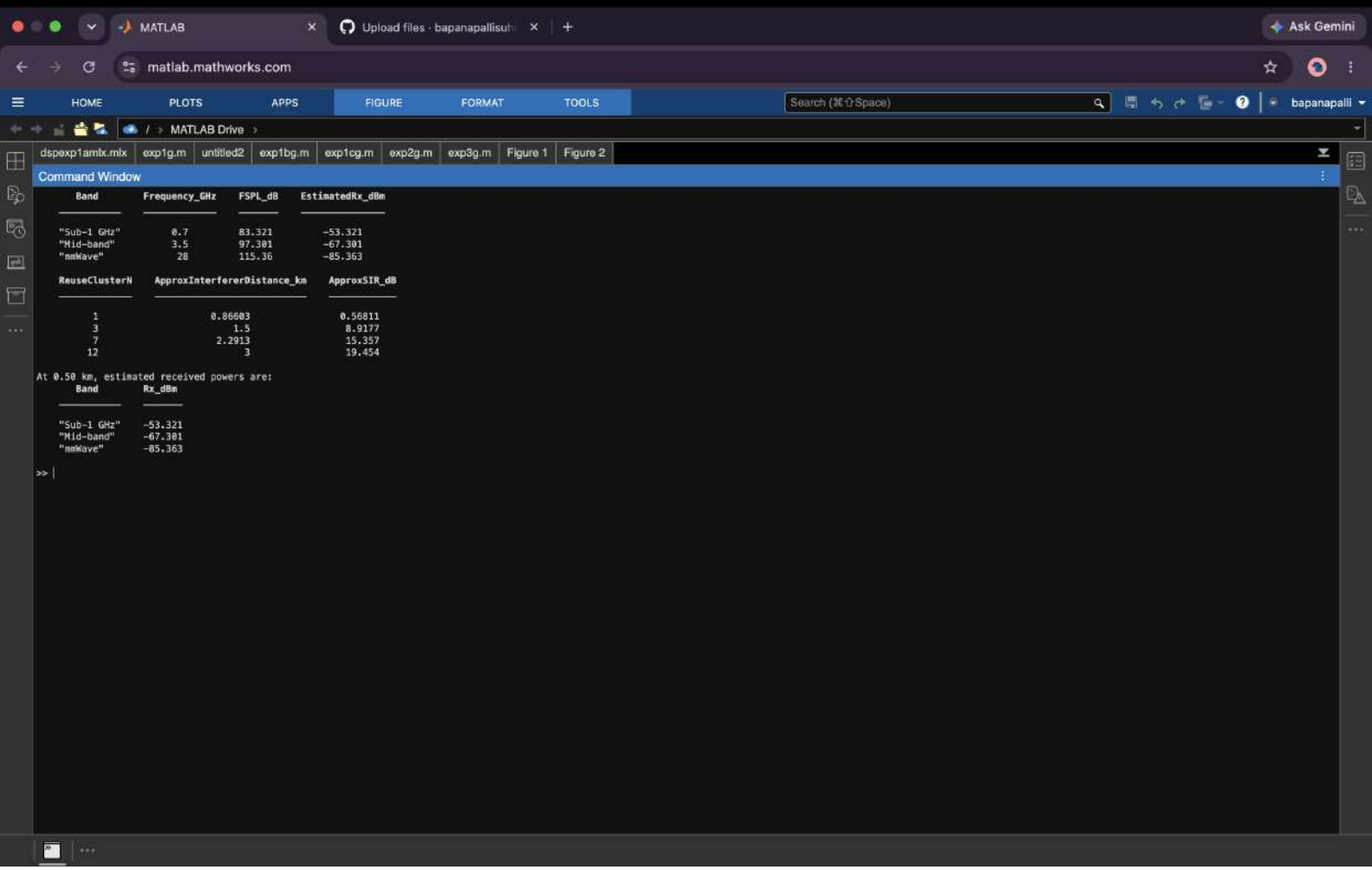
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rxAtPoint = txPowerdBm + gtDbi + grDbi - lossAtPoint;
linkTable = table(bandName,freqGHz(:),lossAtPoint(:),rxAtPoint(:), ...
    'VariableNames',{'Band','Frequency_GHz','FSPL_dB','EstimatedRx_dBm'});
disp(linkTable);
figure; tiledlayout(1,2);
nexttile; hold on; grid on;
for k=1:numel(freqMHz)
    plot(distanceKm,fspl(freqMHz(k),distanceKm),'LineWidth',1.5);
end
set(gca,'XScale','log'); xlabel('Distance (km)'); ylabel('Path loss (dB)');
legend(bandName,'Location','northwest'); title('Free-Space Path Loss');
nexttile; hold on; grid on;
for k=1:numel(freqMHz)
    rx = txPowerdBm + gtDbi + grDbi - fspl(freqMHz(k),distanceKm);
    plot(distanceKm,rx,'LineWidth',1.5);
end
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sirDb = 10*log10(sirLinear);
reuseTable = table(reuseN(:),reuseDistance(:),sirDb(:), ...
    'VariableNames',{'ReuseClusterN','ApproxInterfererDistance_km','ApproxSIR_dB'});
disp(reuseTable);
figure; plot(reuseN,sirDb,'o-','LineWidth',1.5); grid on;
xlabel('Reuse cluster size N'); ylabel('Approximate SIR (dB)');
title('Simplified Co-Channel SIR versus Reuse Cluster Size');
fprintf('At %.2f km, estimated received powers are:\n',distanceAtPointKm);
disp(table(bandName,rxAtPoint(:),'VariableNames',{'Band','Rx_dBm'}));
```

Figures

Figure 1 | Figure 2




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Figure 1
Figure 2
Command Window
Band Frequency_GHz FSPL_dB EstimatedRx_dBm
"Sub-1 GHz" 0.7 83.321 -53.321
"Mid-band" 3.5 97.301 -67.301
"mmWave" 28 115.36 -85.363
ReuseClusterN ApproxInterfererDistance_km ApproxSIR_dB
1 0.86603 0.56811
3 1.5 8.9177
7 2.2913 15.357
12 3 19.454
At 0.50 km, estimated received powers are:
Band Rx_dBm
"Sub-1 GHz" -53.321
"Mid-band" -67.301
"mmWave" -85.363
>>